Pranav V

+91-7676858328 | pranavv736@gmail.com | linkedin.com/in/pranav-venu-550729264 | github.com/pranavv1210 | Bengaluru, India

Summary

Machine Learning Engineer specializing in **computer vision and deep learning**, with hands-on experience building production-grade AI systems across image segmentation, model optimization, and real-time inference. Strong foundation in Python, data pipelines, and applied research, with a track record of shipping optimized, deployable solutions from prototype to production.

Experience

Cheerans Global Solutions Pvt. Ltd. Jan. 2026 – May 2026
Data Scientist Intern Bengaluru, India

- Engineered a **data preprocessing and feature-engineering pipeline** across multiple business datasets, cleaning and structuring raw data to support downstream analytics
- Designed and deployed **interactive Power BI/Tableau dashboards** to surface real-time KPIs, enabling data-driven decision-making for stakeholders
- Built a **component-based frontend architecture using React.js** for a café billing software system, implementing state management and responsive UI for production use

Projects

CardioGuard: AI-Powered Mobile Heart Disease Analyzer | *Python, TensorFlow, TFLite, Librosa, OpenCV*

- Built a deep learning heart-sound classifier on the PhysioNet 2016 dataset (3,240+ recordings) using mel-spectrogram image preprocessing, achieving **92.23% AUC** accuracy
- Optimized model with TensorFlow Lite, **reducing size by 91%** (1.33MB → 0.12MB) for real-time, on-device inference
- Deployed a cross-platform **Streamlit web app** supporting QR-based workflows and offline mobile-first screening

Operation Gridlock: Gen-AI Vehicle Detection & Pursuit System | *SAM3, Stable Diffusion, LLaMA 3.2, Computer Vision*

- Engineered a real-time **computer vision pipeline** for CCTV vehicle tracking using SAM3 segmentation-level detection and Real-ESRGAN/Stable Diffusion image restoration for low-quality footage
- Built a **route-forecasting engine** predicting suspect vehicle trajectory and ETA, triggering automated geo-alerts across a multi-camera network

AI Resume Screening System with Chatbot | *React.js, Node.js, Hugging Face, Supabase, NLP*

- Developing a full-stack platform using **NLP-based semantic matching** to score and rank candidates against job descriptions with explainable, bias-mitigated outputs
- Integrated an **AI chatbot** (Rasa) delivering real-time, explainable feedback to candidates pre-submission

Education

CMR Institute of Technology 2022 – 2026
Bachelor of Engineering, Artificial Intelligence & Machine Learning CGPA: 8.66

St. Joseph's Pre-University College 2020 – 2022
Pre-University Course – Science (PCMB) 83.3%

Cambridge School 2018 – 2020
10th Grade – ICSE 81.8%

Technical Skills

Machine Learning & AI: Deep Learning, Computer Vision, NLP, Model Optimization, TensorFlow, Keras
Gen-AI Tools: OpenAI, Generative AI, LLMs, RAG
Programming Languages: Python, Java (Intermediate), SQL (Intermediate)
Frameworks: React.js, Node.js
Web Technology: HTML, CSS
Cloud Platforms: AWS (Basic)
Software & Analytics: Visual Studio Code, Jupyter Notebook, Power BI, Tableau, Android Studio, MS Excel
Version Control: Git, GitHub
Operating Systems: Windows

Achievements & Extra-Curricular

- Secured **2nd Place** in the National Level Deep Learning Hackathon organized by BGSIT (2025)
- Filed a patent for an **AI-Powered Pet Monitoring System** using computer vision for real-time health alerts
- Participated in Gen AI Hackathon (Trendy Tech), Ops-Fusion Hackathon, and Vyuhatech 2.0, all at CMRIT (2025)
- Served as **Class Representative** for 1st and 2nd semesters; Member of Cybernauts AI/ML Club